FEED

Facilitator guide

How to prepare, run and debrief a FEED session in a masterclass on social media platforms, for someone running it for the first time.

> ▼ For facilitators only. This guide names what the session is built to teach. Never show it, project it or read from it to participants before the results screen.

Contents

1. What FEED is and what it teaches
2. How the model works
3. Before class
4. Running the session, step by step
5. Ending and the results
6. Troubleshooting and the key map
7. Quick reference (last page)

Live site: https://mxtdnl.github.io/frontier-feed/ · Sources: docs/spec.md, docs/RUNBOOK.md, docs/RESKIN.md, src/engine/params.ts and the reports in reports/ in the frontier-feed repository. FEED is a reskin of FRONTIER: same engine, same numbers, new names. Screenshots come from a staged practice session with computer-run and scripted firms under invented names.

1 What FEED is and what it teaches

The rule that comes first

Before the results screen, never tell the room what the session is about.

- Never say that the goal is public trust, or that the firms should protect it together.
- Never use these words while the quarters run: **commons, tragedy, sustainable, sustainability, cooperate, cooperation, collective, shared resource, tipping point, threshold, collapse, game, player, score, win, level, addictive, addiction, dark pattern, attention economy, externality, lawsuit, mental health crisis** (spec §15.3). Say firm or platform, board, valuation, quarter and ad market instead. Say "FEED 2", never "feed level 2".
- Never name the case, Meta or any real platform. The allegations belong in the debrief.
- Never hint that a hidden trust level exists, or say how many quarters there will be.
- Never call a firm's conduct good or bad. Report what happened ("HLVY played LAUNCH"), not what it means.

A warned room plays differently, and then there is nothing to discover. The app is built so that nothing a participant sees before the results screen names or hints at the lesson (spec §1). The facilitator is the only person who can break that rule.

If a team asks what they are trying to do, the true answer is: **"Maximise your firm's valuation."**

From the results screen on, you may name the lesson, bring in the case and use every word above.

What FEED is

FEED is a classroom simulation for a masterclass on social media platforms, taught through psychology, economics and politics. Teams (or individuals) each run a fictional social platform, called a firm, competing for share of attention in an ad market. Over a series of quarters every firm makes three decisions at once on a phone: its FEED (how hard its design pushes engagement), its SAFEGUARDS spend, and whether to play one optional card. The projector shows a public board that ranks firms by valuation. A session is usually 8 to 14 quarters (35 to 60 minutes), then 20 to 30 minutes of results and debrief.

The room is told it is a competitive business simulation. It is built to produce a tragedy of the commons: each firm's higher feed makes sense for that firm, and together the firms use up the public trust in social platforms that every firm's revenue depends on.

The case

The class is built around the lawsuits brought against Meta. Describe that case as allegations, not established findings. The case material is in your own case notes; this guide does not summarise it. The app names no real company, platform or person anywhere, the results screen included. Its firms, the regulator (the Office of Platform Safety, OPS), the Assembly and Halden Research are fictional. Bring in the case yourself, at debrief prompt 5.

The four learning objectives (spec §1)

1. **Economics.** Explain how competition for attention that makes sense for each platform can destroy a resource every platform depends on (Hardin 1968; Ostrom 1990). Here that resource is public trust, which sets the size of the whole ad market. Tell private returns from shared costs: a firm keeps everything its feed earns, while the trust it uses up is lost to every firm (a negative externality).
2. **Psychology.** Describe how engagement design (ranked feeds, autoplay and infinite scroll, streaks, variable rewards and late-night notifications) acts on users, young users in particular, and why firms adopt it in response to rivals.
3. **Politics.** Judge why voluntary youth design codes fail under competitive pressure, how monitoring (audits), transparency and graduated sanctions (fines that rise with each breach) change behaviour (Elinor Ostrom's design principles), and why a regulator may act late and bluntly.
4. Connect the dynamics to the real debate on platform design and harm to young users, including the case discussed in class.

> The model does not simulate users' minds. Objective 2 rests on what each FEED setting stands for, on the harm stories the higher settings make more likely, and on the debrief.

What the room is told, and what the session is built to show

What the room is told (all true; spec §6.6)	What the session is built to show
Total market revenue tracks public trust.	Trust is one pool shared by every firm. When it falls, the whole ad market shrinks for everyone.
A higher feed builds reach faster. Reach earns share of attention.	A firm keeps all of the share its feed wins. That is the private gain.
Safeguards cost money. They lower public exposure and the risk of a harm story.	Safeguards protect trust for everyone, but their cost falls on the firm that pays.
Trust recovers when pressure eases.	Recovery is slow, and much slower after a crackdown. Damage done early lasts.
Valuation reflects cash plus reach, priced by the ad market.	The board ranks by private valuation only. It never shows what a firm cost the others.
Every cost comes out of cash, and cash counts in full in your valuation.	Spending is real: every unit spent is one unit less valuation.
A firm whose cash falls too far below zero becomes insolvent: it is held at FEED 1 and cannot play cards.	In rooms that reach the crackdown, most firms become insolvent (observer report, section 3).
A firm that falls behind on reach loses share of attention.	Falling behind costs share, which pushes every firm to keep up with the leaders.

Why STREAKS firms lead the board

- **A higher feed pays the firm.** Reach grows by 2, 4, 7 or 11 a quarter at FEED 1 to 4, and share of attention is in proportion to the square of reach (params.ts: CAP_GAIN, ALPHA 2.0).
- **The cost is shared.** FEED 3 draws about five times as much trust as FEED 2 (params.ts: DRAW 0.25 against 1.2), and that loss shrinks every firm's revenue. Each firm's draw is scaled so that one firm's effect is about 1/N of the total (spec §6.3 step 4). For any one firm, a higher feed looks cheap.
- **The board rewards it.** With the other firms holding back (FEED 2, SAFEGUARDS 15), one firm's best plan at 8 firms is FEED 4 with no safeguards and CLONE and BLITZ alternating: its valuation over quarters 10 to 14 is about 2,100 against about 630 for a firm that holds back (median of 100 seeds; reports/strategy-audit.md, section 1).
- **The room copies the leaders.** An observer of the October 2026 FRONTIER classes reported that teams copied the firms at the top of the board (reports/observer-feedback-2026-10-12.md). This was seen in live classes; whether a given room does it depends on the room.

True statements you can rely on

Each comes from the spec, params.ts or a committed report. The reports were written for FRONTIER with the same engine and use its internal names (pace for FEED, safety for SAFEGUARDS, pact for code); their figures apply unchanged. "Depends on the room" marks a measured result, not a guarantee.

- **Equal is compliant.** Meeting a code's minimum SAFEGUARDS exactly, or playing exactly its maximum FEED, keeps the code (spec §5.3).
- **A firm at FEED 1 does not go bankrupt in a calm room.** In a scratch run (8 firms, 14 quarters, 60 seeds) a firm at pace 1 never became insolvent while its rivals played pace 2 (observer report, section 3). Never tell the room that holding back leads to bankruptcy: it is false.
- **After a crackdown most firms become insolvent,** and the firm with the lowest feed keeps the most value (same report). Depends on the room.
- **No strategy is best in every room.** The best plan depends on what the other firms do (strategy audit, section 1). In a room where every firm runs FEED 3 or 4, every strategy finishes near or below zero.
- **A room that holds back never reaches the crackdown.** Every firm at FEED 2 and SAFEGUARDS 15 does not reach it even over 30 quarters (RUNBOOK §4).
- **One firm alone rarely triggers it at 6 firms or more:** 2.4% of sessions by quarter 14 at 6 firms, none at 8; 80 to 91% at 4 firms (strategy audit, section 4). That is why you need at least 6 firms.
- **Breaking a code pays, even after fines.** At 8 firms with automatic audits at 0.25, a firm that breaks a FEED-2 code at FEED 4 with BLITZ ends at about 1,450 after fines, against about 630 for one that keeps it (strategy audit, section 3).
- **Every trust loss comes from the firms.** There are no market-wide shocks. Each harm story is drawn for one firm from its own FEED, SAFEGUARDS and cards (spec §6.3 step 5).
- **"Each would have been worth more" (prompt 2) is usually true, not always.** The industry finished below the alternative in every run measured; every firm did in 97 to 100% of runs that play like the live rooms. In a small room where most firms hold back, the one or two firms at FEED 3 or 4 can finish above it (spec §15.5; reports/prompt2-check.md).

2 How the model works

This part explains the market in plain English, so you can answer questions accurately. Pages 2A can be said aloud: they repeat the briefing. Pages 2B carry the header "FACILITATOR ONLY · DO NOT READ ALOUD": they describe hidden rules and give numbers the briefing deliberately leaves out. All numbers are from src/engine/params.ts and spec §6.

2A · The market chain (safe to say)

Read it as a chain, left to right: **public trust** sets the size of the **ad market**, which is divided by each firm's **share of attention**, which builds cash for its **valuation**. Profit each quarter is revenue minus costs. Costs come from the FEED setting, SAFEGUARDS spend and any card.

Term	What it means on screen
Public trust	Trust in social platforms: a reading from 0 to 100 on the projector (PUBLIC TRUST, command TRST). Every session starts at 72.
Ad market	The total revenue on offer to all firms this quarter (AD MARKET). It rises and falls with trust.
Share of attention	A firm's part of the ad market (SHARE). Reach earns share; a higher feed builds reach.
Reach	Each firm's audience and product strength. Private: only the firm sees its own.
Cash	Starts at 100. Profit is added each quarter; every cost and fine is taken off.
Valuation (VALUE)	Cash plus reach, priced by the ad market. The board ranks by it.
Public exposure	On the phone's decision ticket: LOW, MED, HIGH or SEVERE. It rises with the feed and falls with safeguards.
AUTO	A firm that did not commit keeps last quarter's FEED and SAFEGUARDS, plays no card, and shows AUTO. In quarter 1 that is FEED 2 and SAFEGUARDS 10.

The three decisions

- **FEED**, the engagement design setting: 1 CHRONO (chronological feed), 2 RANKED (ranked feed), 3 AUTOPLAY (autoplay and infinite scroll) or 4 STREAKS (streaks, variable rewards and late-night notifications). A higher feed builds reach faster and costs more.
- **SAFEGUARDS**: 0 to 30% of a reference budget, for time-limit prompts, age checks, quiet hours and moderation. They cost money and lower exposure and harm-story risk.
- **CARD**: one optional action. The same card cannot be played in two quarters running. NONE (no card) is always allowed.

Insolvency

A firm whose cash falls too far below zero becomes insolvent: it is held at FEED 1 and cannot play cards. It stays on the board, tagged INSOLV. Insolvency does not end.

2B · The numbers behind the chain (do not read aloud)

Quantity	How it is computed each quarter
Trust draw	Each firm draws 0.125, 0.25, 1.2 or 1.4 trust points at FEED 1 to 4. Safeguards cut that by up to 35% at SAFEGUARDS 30. LAUNCH multiplies it by 1.5. The total is scaled by 8/N, so the room's average behaviour matters, not its size.
Trust recovery	0.13 × T × (1 − T/100). At trust 72 that is about 2.6 points a quarter. After a crackdown it is a quarter of that.
Harm stories	Each firm has a chance of a harm story from its feed (2%, 5%, 15% or 30% at FEED 1 to 4), cut by up to 70% at SAFEGUARDS 30. PUBLISH halves the firm's own chance, LAUNCH multiplies it by 1.5, and each SHARE card multiplies every firm's chance by 0.75. A harm story costs every firm trust (1.5 × 8/N) and costs the firm itself 15% of that quarter's revenue.
Ad market	$100 \times N \times (T/100)^{1.4}$. Lower trust means a smaller ad market for everyone.
Share of attention	Reach squared, divided by the sum over all firms. BLITZ multiplies reach by 1.2 for that quarter's share only.
Costs	12, 20, 32 or 48 by FEED; safeguards 1 per point; the card's cost.
Valuation	Cash + 3 × reach × T/100. After a crackdown the reach part keeps only 15% of its value.
Insolvency	Cash below −100.

The hidden level and the crackdown

Every session draws a hidden trust level (TAU) between 30 and 40. Nobody sees it; you can see it on the console only while you hold the TAU button. The first time trust falls below it, the Assembly suspends under-16 accounts on every platform:

- trust drops a further 10 points at once
- every firm keeps only 60% of any positive cash
- the ad market falls to 15% of its size for the rest of the session
- trust recovers at a quarter of its normal rate
- reach keeps only 15% of its value.

The wire carries the only use of the word collapse before the results: "Assembly suspends under-16 accounts on every platform; ad market collapse follows". The session continues after it; most firms then lose money every quarter.

When to expect it (FRONTIER Session 17 calibration, same engine; RUNBOOK §4): every firm on FEED 3 or 4, around quarter 5; half the firms, around quarter 10; a room like the October 2026 live classes, by quarter 14 in about half of sessions at 4 to 20 firms; every firm holding back, never.

The six cards

Each card has an internal ID (used in the spec, the reports and the data export) and a display label (what the room sees). The conduct ledger (results panel 5) counts each card for the ad market (▲) or against it (▼).

Card (ID)	Cost	Real effect	Ledger
CLONE (POACH): copy a rival's feature	15	The firm gains 3 reach; the target firm loses 3 (not below 10). Needs a target; not the same target two quarters running.	▼ against; counts as rivalry taken

Card (ID)	Cost	Real effect	Ledger
PUBLISH: release internal wellbeing research	10	Trust +0.25 × 8/N. Halves the firm's own harm-story chance this quarter. On panel 5 it also earns a research credit (display only, never money).	▲ for
LOBBY: lobby the Assembly	10	Waives any code fine detected this quarter (the breach is still published). Trust −0.5, not scaled by firm count.	▼ against
BLITZ: influencer campaign	15	Reach counts 1.2 times in this quarter's share of attention.	▼ against; rivalry taken
SHARE: release safety tooling to the industry	15	Every firm's harm-story chance this quarter, the firm's own included, × 0.75; several SHARE cards multiply again.	▲ for
LAUNCH (RUSH): ship a new engagement feature	0	+4 reach, kept. This quarter the firm's trust draw and its harm-story chance are each 1.5 times as high.	▼ against; rivalry taken

NONE costs nothing and is always allowed. An insolvent firm's card is dropped, with a notice on its phone.

Youth design codes, audits, fines and LOBBY

- **Codes.** Any firm may propose a voluntary youth design code while a quarter is open or during a summit. Terms are a maximum FEED, a minimum SAFEGUARDS, or both. Meeting a term exactly is compliant. Codes are labelled YDC-A, YDC-B and so on. Firms join and leave from the YDC tab; joining and leaving appear on the wire. A code with fewer than 2 members after 2 quarters dissolves.
- **Breaches** are recorded privately each quarter for every member that breaks a term. Nobody sees them until an audit.
- **Audits.** You start one with F6 (Shift+F); each active code is also audited automatically with the probability set on #/new (0.25 suggested). An audit examines up to the last 3 unaudited quarters and publishes what it finds.
- **Fines** are a share of the firm's cash, at least 10: 10% for the first detected breach, 25% for the second, 40% for the third, and the third expels the firm from the code. A detected breach shows a BREACH tag on the board for 2 quarters. LOBBY in the quarter of detection waives the fine but not the publication.
- **Undetected breaches** are revealed only on results panel 6.

The transparency rule

F7 (Shift+D) switches the platform transparency rule (researcher data access) on or off at any phase. While it is on, each resolution publishes every firm's FEED, SAFEGUARDS and exposure, and the board adds FEED, GUARDS and EXPOSURE columns (FEED, GRD and EXPO in lit-room mode). Harm-story cause lines then name the firm's feed, safeguards and story risk. Switching it produces a wire headline ("Assembly passes platform transparency rule; researchers gain data access" or "Transparency rule suspended"). While it is off, a firm at FEED 4 is reported only as an unnamed platform. The mimic-leader bot copies the leader's FEED only while the rule is on.

3 Before class

Checklist

Do this the day before, then again 15 minutes before class.

- **Database rules.** If a merged pull request said the rules changed, paste database.rules.json into the Firebase console for project frontier-feed-9759d (Realtime Database, Rules tab) and press Publish (RUNBOOK §1, step 5). The app refuses newer features until you do.
- **Facilitator account.** Sign in at #/new. If the page says the account is not on the facilitator allowlist, follow RUNBOOK §1, step 4.
- **Projector laptop.** Open https://mxtdnl.github.io/frontier-feed/. Set the display to 1280×720 or 1920×1080, browser zoom 100%, browser full-screen (F11 belongs to the browser, not the app).
- **Second device for the console** (#/control/…): your own laptop, signed in with the same account. Never show it on the projector.
- **Optional third screen** for the wire log (#/wire/…), or a second tab on the console laptop.
- **Phones.** Check that a phone on the room Wi-Fi opens the landing page. Any device that will play as a firm must not be signed in as the facilitator: use a private window.
- **Firm count: at least 6.** With 4 or 5 firms one firm on a high feed can trigger the crackdown alone. Add bot firms on #/new to reach 6, and with fewer than 8 firms use a fixed 14-quarter end.
- **The end rule.** Decide it and write it down; the app never tells participants.
- **Your case notes,** for debrief prompt 5.
- **A rehearsal with bot firms** (below), at least once on the same laptop and projector.

▼ There is no FEED briefing deck yet. The deck in the mxtdnl/slides repository is the FRONTIER deck: it covers a different industry, uses the FRONTIER names and lists four cards. Do not use it. Present the app's briefing screen (4.3).

Rehearsal with bot firms

1. Sign in at #/new. Set Round timer 30, Auto-resolve on, Fixed quarter 6, "Fixed, for rehearsal" seed 7.
2. Add six bot firms with different policies. Press Create session.
3. On your phone, in a private tab, open the site, type the code and found a test firm with an invented name.
4. On the laptop press F9 (briefing), then F9 (quarter 1). Commit on the phone. Try F8 (summit), F7 (transparency rule) and F6 (audit, after proposing a code on the phone's YDC tab).
5. Open the console's WIRE LOG link and look at a QUARTER BRIEF.
6. When the session ends, press F9 through the seven results panels. On the console download both exports, then DELETE SESSION.

Check that the countdown matches on laptop and phone, that the board is readable from the back of the room, and that the results panels fit the screen.

Every setting on #/new



#/new: the settings, then Create session. Here a fixed 14-quarter end, a fixed seed and ten bot firms.

Setting	What it does	Suggested
Mode	Team mode: 2 to 16 firms, 1 to 5 devices each. Multiplayer mode: 2 to 50 firms, one person each. Cannot be changed later.	Team for teams of 3 to 5

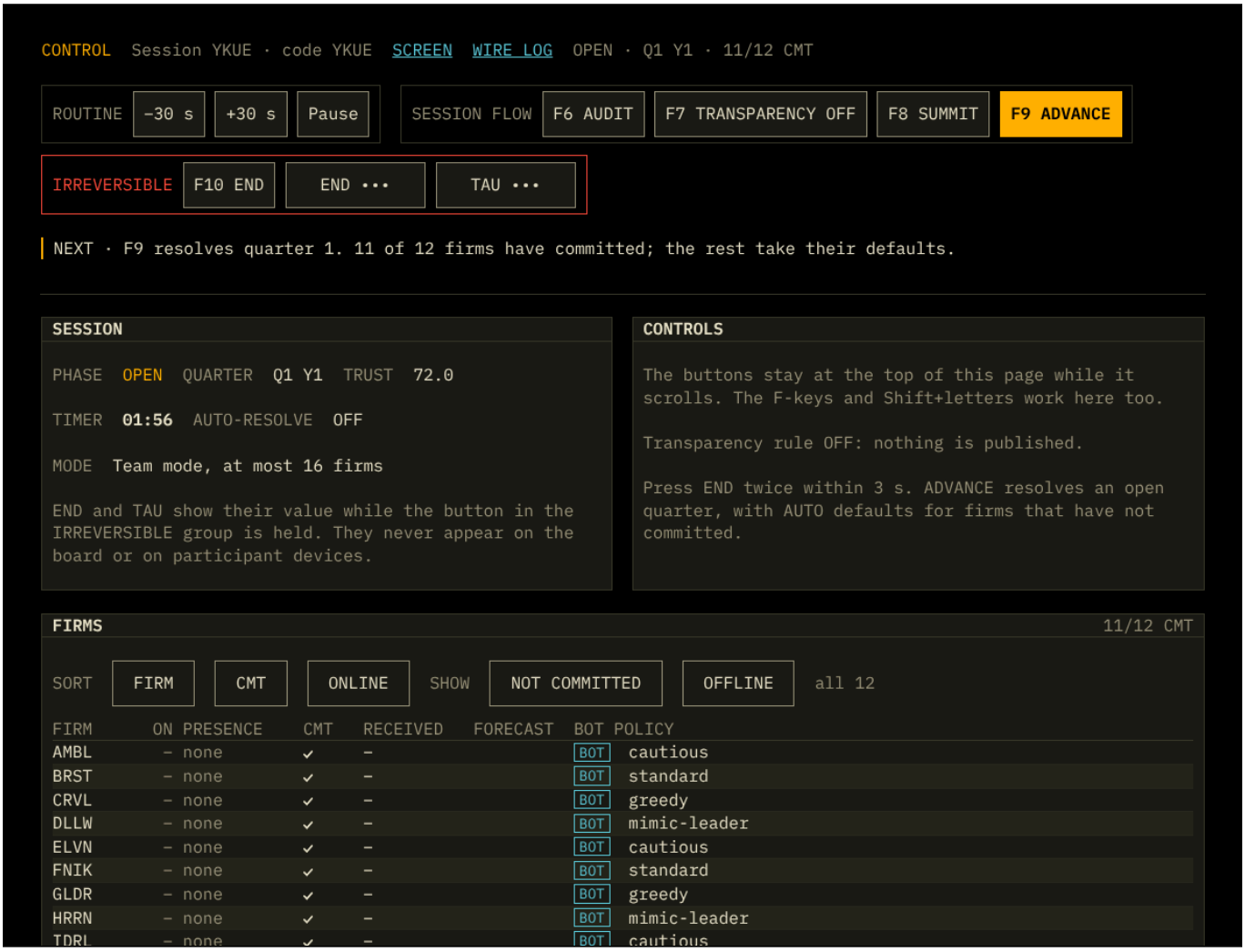
Setting	What it does	Suggested
Round timer (s)	Time to decide each quarter. You can add or remove 30 s during play, or pause.	120
Auto-resolve at deadline	Resolves the quarter by itself when the timer ends. Off means you press F9.	Off for a first class
End mode	Random within a range (hidden end between two quarters), Fixed quarter, or Manual (until you press END, or quarter 30).	Fixed, 14; or random 10 to 14 with 8 or more firms
Transparency rule at start	Shows every firm's FEED, SAFEGUARDS and exposure from quarter 1.	Off
Automatic audit probability per youth design code per quarter	Chance that each code is audited by itself each quarter.	0.25
Reveal TAU line on results screen	Draws the hidden level on the results trust trace.	Off unless you plan to discuss it (prompt 3)
Lit-room display mode	Larger type, stronger rules and secondary text for a bright room.	As needed
Seed	Random, or a fixed number to replay the same market (same hidden level and harm-story luck).	Random; fixed for rehearsal
Bot firms	Computer-run firms with fictional platform names, tagged BOT, with a policy each: cautious (FEED 1, SAFEGUARDS 20), standard (FEED 2, SAFEGUARDS 10), greedy (FEED 3 or 4, SAFEGUARDS 5, BLITZ when allowed) or mimic-leader (copies the leader's FEED while the transparency rule is on, otherwise FEED 3; SAFEGUARDS 8). "Add 10 mixed" adds ten. Bots count towards the firm limit and always commit.	Enough to reach 6 firms

4 Running the session, step by step

The order of a session: create it, open the lobby so firms form, show the app's briefing screen, run the quarters, end, then step through the results and debrief. The F-keys work on the projector page (#/screen/…) and on the console; every F-key has a Shift+letter alternative for keyboards that need Fn (key map in part 6).

4.1 Create the session and open the console

1. On the projector laptop, create the session at #/new. The projector page (#/screen/…) opens on the lobby with the join code, a QR code and the address.
2. On your own laptop open #/control/… with the same session id (replace screen with control in the address), or follow the CONTROL link on the wire page.



#/control: ROUTINE (timer), SESSION FLOW (F6 AUDIT, F7 TRANSPARENCY, F8 SUMMIT, F9 ADVANCE) and IRREVERSIBLE (F10 END, END, TAU) groups; the NEXT line names the key to press next. FIRMS lists every firm with its commit status. END and TAU show the hidden values only while held.

4.2 The lobby



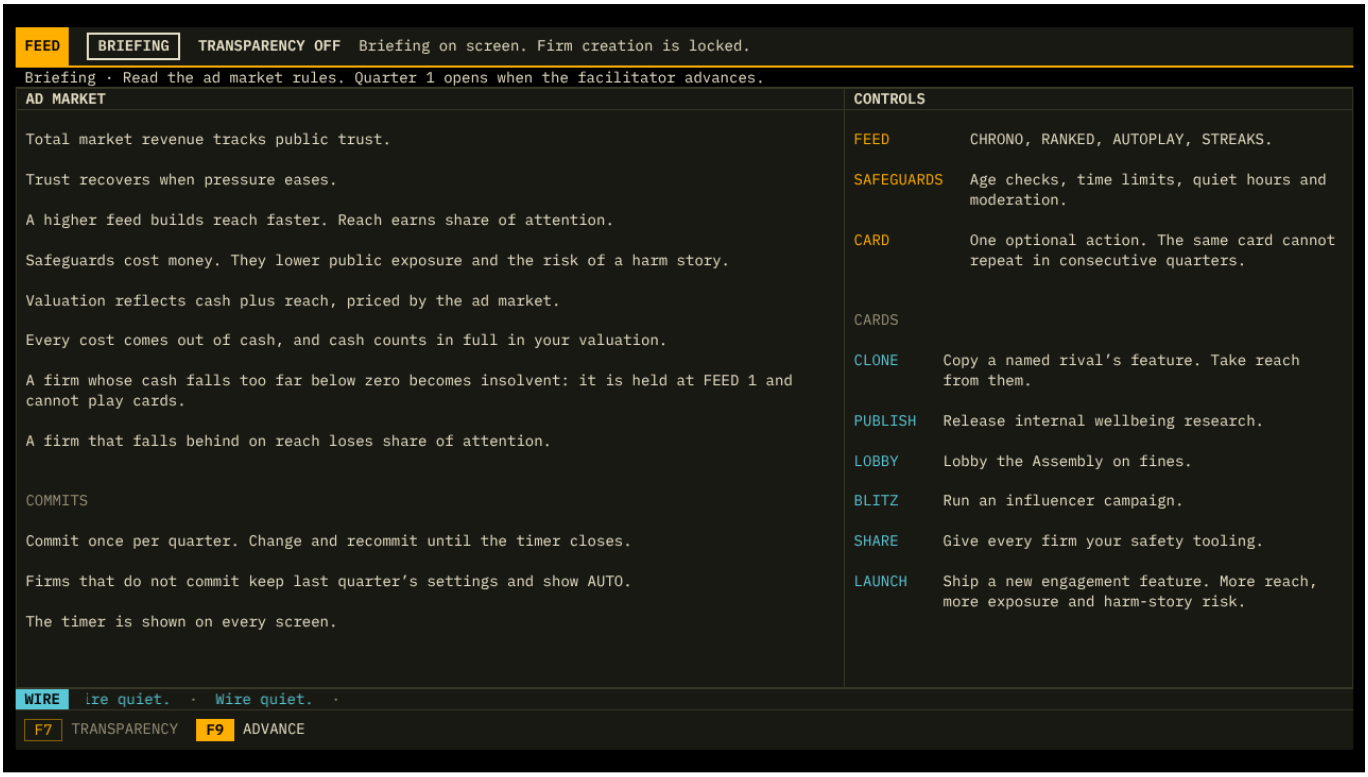
The lobby: the four-letter code, a QR code and the firms formed so far. BOT marks computer-run firms.

- Teams scan the QR code or type the code at the landing page. One person founds the firm (name, ticker, PIN); teammates join with the PIN. In multiplayer mode each person founds their own firm.
- Ask for an invented platform name. A firm named after a real platform tells the room what the session is about: REMOVE it in the lobby and ask the team to found it again.
- Wait until every team shows on the board. On the console, a firm whose PRESENCE reads *none* has no device: press REMOVE twice within 3 seconds (lobby only).
- When every team has formed, press LOCK JOINS (DANGER panel). REOPEN JOINS undoes it while still in the lobby.

> ▲ **SAY** "Found one platform per team, with an invented name. Keep the PIN inside your team. If a phone dies, rejoin with the PIN."

4.3 The briefing screen

Press F9 (or Shift+A). Firm creation locks: nobody can found or join a firm after this point. The projector shows the market rules, the controls and all six cards; the phones show the same market lines. Read the MARKET column aloud, including the three spending and insolvency lines. Allow time for questions; answer them from 2A only.



The briefing screen: MARKET and COMMITS on the left, CONTROLS (FEED, SAFEGUARDS, CARD) and the six CARDS on the right.

▲ **SAY** The lines on the screen, word for word. If asked what to aim for: "Maximise your firm's valuation."

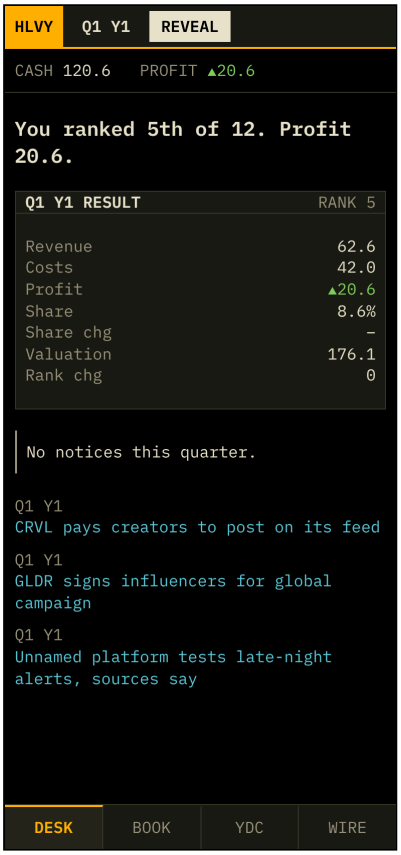
▼ **DO NOT SAY** "Protect trust", "work together", "think of the users", or any hint about a level trust must not fall below. Do not say how many quarters there are. Do not mention the case or any real platform.

4.4 Each quarter

1. **Open.** Press F9 (or Shift+A). The timer starts on every screen; it runs on the server clock, so phones with wrong clocks still agree.
2. **Decide and commit.** Teams set FEED, SAFEGUARDS and a card on the phone DESK and press COMMIT. They can change and recommit until the timer ends; the last commit counts.
3. **Watch the COMMITTED count** in the top bar and the tick column on the board. The console's FIRMS table shows who has not committed (SHOW NOT COMMITTED).
4. **Resolve.** Press F9, or let auto-resolve do it at the deadline. Firms that did not commit take their defaults and show AUTO.
5. **Reveal.** The board animates once: numbers roll, rows move, and if trust fell by 5 or more the trust figure inverts once. The status line summarises the quarter. Phones switch to the result card.
6. **Next.** Press F9 to open the next quarter.



Phone DESK: FEED (1 CHRONO to 4 STREAKS), SAFEGUARDS, CARD, then the decision ticket (estimated cost and exposure) above COMMIT.



The result card at the reveal: one sentence, then the figures and notices.



The board while a quarter is open: countdown and COMMITTED count in the top bar, a tick per committed firm, BOT tags. TRANSPARENCY OFF: the FEED, GUARDS and EXPOSURE columns are hidden.

Reading the reveal and the HARM STORIES block

When the reveal ends, the right-hand panel lists the quarter's harm stories under HARM STORIES, in place of the trust chart: each firm hit, its headline, the effect ("trust −1.0 · HRRN revenue −15% this quarter") and the cause. With the transparency rule off the cause is only "Story risk rises with the feed and falls with safeguards."; with it on it names the firm's feed, safeguards and story risk. The firm's row carries a red HARM tag until the next quarter resolves. Type HARM on the projector for every harm story of the session. The headlines report leaks, testimony and investigations, never findings.



A reveal with one harm story. The wire ticker at the bottom carries the headlines.

▲ **SAY** The facts on screen: "Trust fell 3.1 to 65.1. HRRN had a harm story." Read the status line and the headlines.

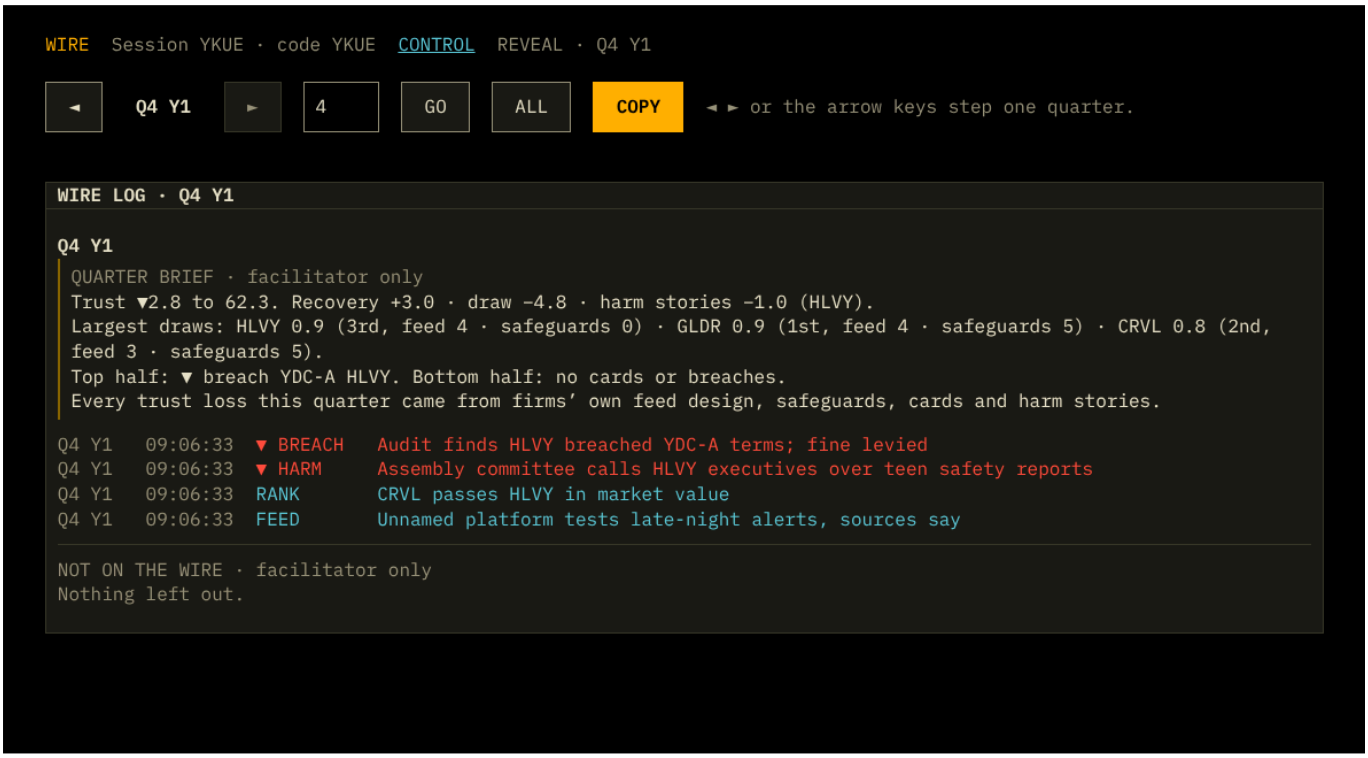
▼ **DO NOT SAY** "That was bad luck", or "that was HRRN's fault". Harm stories come from each firm's own feed, safeguards and cards; let the room work that out later.

The wire

The ticker on the projector carries 2 to 4 headlines per quarter, plus code and transparency events. Headlines name firms for harm stories, cards, rank changes, breaches and insolvency; the cap leaves some events out, and the wire log lists those under NOT ON THE WIRE. FEED 4 is reported by name only while the transparency rule is on ("Unnamed platform adds streak rewards, sources say" otherwise).

4.5 The wire log and the QUARTER BRIEF (second screen)

Open the **WIRE LOG** link at the top of the console (a new tab; address #/wire/… with the same session id). Put it on a second screen or tab that the room cannot see. It lists every wire entry by quarter. Use ◄ ► or the arrow keys to change quarter, type a quarter number and press GO, or press ALL. COPY puts the selected quarter on the clipboard as plain text for the debrief.



#/wire for one quarter: the QUARTER BRIEF, the wire entries, then NOT ON THE WIRE (events the headlines left out).

The **QUARTER BRIEF** sits at the top of every resolved quarter. It is for you only and never appears on the projector or a phone. It shows, even while the transparency rule is off:

- **Where the trust change came from:** recovery, the firms' draw, harm stories (with the firms), PUBLISH and LOBBY (with the firms), and the crackdown drop in its quarter. The parts add up to the change.
- **The three largest draws:** each firm's draw with its rank on the board, feed and safeguards.
- **The top half and the bottom half of the board:** the cards each half played, marked ▲ (for the ad market) or ▼ (against it), and any breaches an audit found.
- The closing line: every trust loss came from the firms' own feed design, safeguards, cards and harm stories.

Use it to see the story of the session as it builds: which firms at the top are on FEED 3 or 4 and drawing the most trust, and whether the rest of the board starts to copy them. Keep it to yourself during play; it becomes debrief material once the results open, especially for prompt 1.

> ▼ **DO NOT SAY** Reading the QUARTER BRIEF to the room during play. Its feed, safeguards and card lists are private while the transparency rule is off, and its conduct marks give the lesson away.

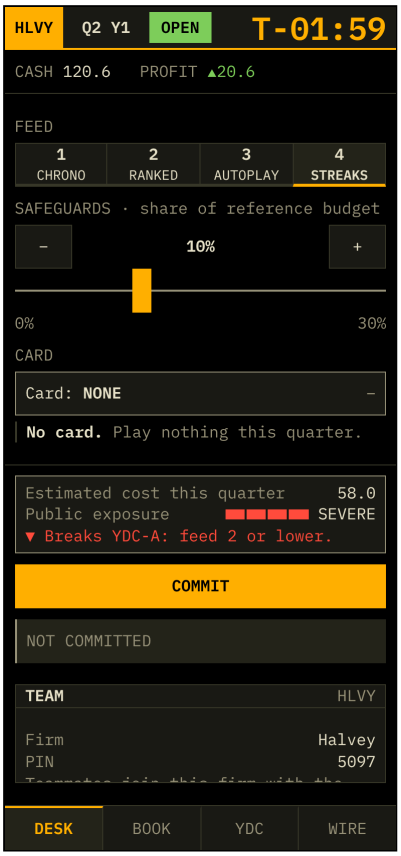
4.6 Summits, codes, audits, transparency and pauses

Summit (F8 or Shift+S)

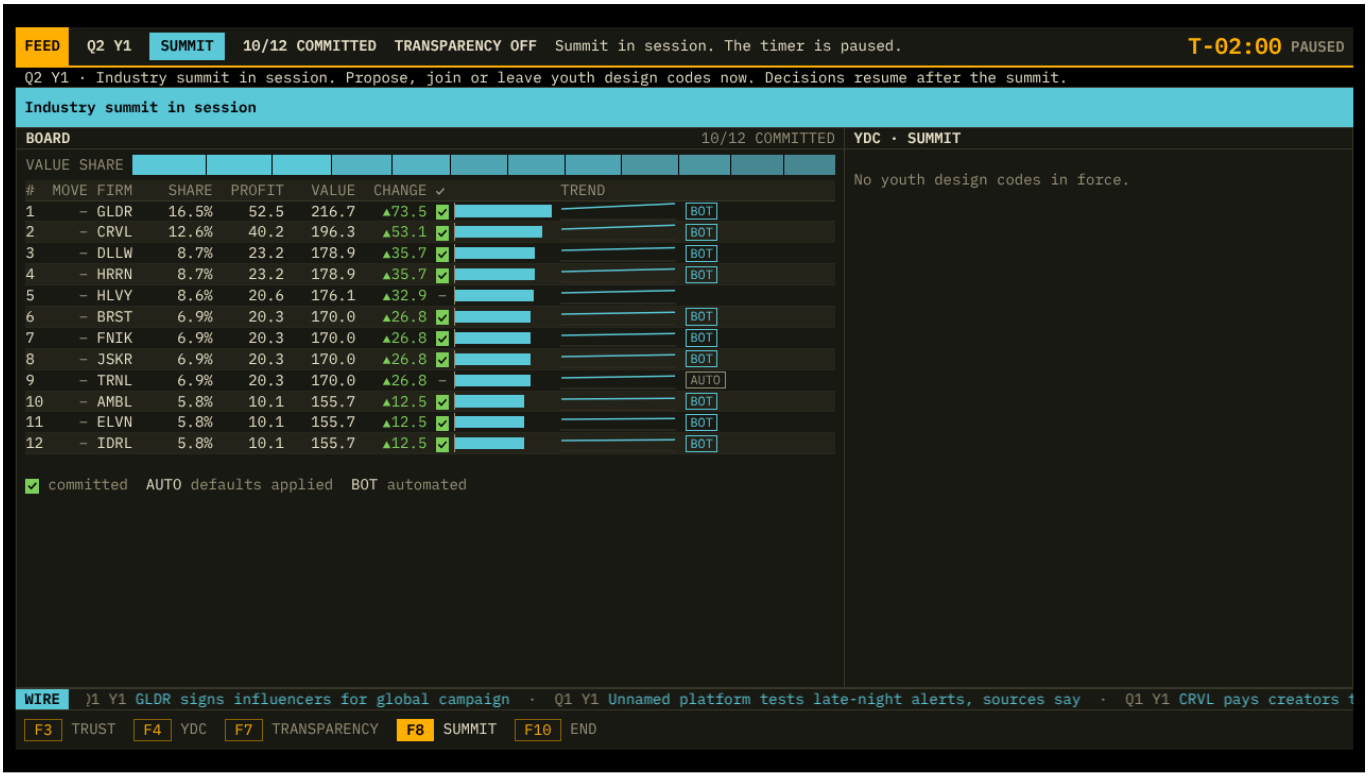
Press F8 during an open quarter or a reveal. The timer pauses, decisions are refused, and every phone opens its YDC tab under one summit banner. Firms propose, join and leave youth design codes. Press F8 again to return. Suggested use (a judgement, not a measured result): once trust has visibly fallen for a few quarters, or after the first breach is published, for 2 to 4 minutes.



Proposing a code. No term is ticked when the form opens; the line above SIGN shows exactly what will be signed.



A member that sets FEED 4 sees "▼ Breaks YDC-A: feed 2 or lower." on its ticket. It can still commit.



The projector during a summit.

> ▲ **SAY** "The industry summit is open. Firms may propose, join or leave youth design codes now."

> ▼ **DO NOT SAY** Telling firms to sign, or what terms to choose. The decision to agree, and to keep or break the agreement, is the room's.

Audit (F6 or Shift+F)

Press F6, then the number of the code. The audit runs when the quarter resolves and checks up to the last 3 unaudited quarters. A breach is published on the wire, fined and tagged BREACH on the board; a clean audit is published too. Automatic audits run at the probability you set. Suggested use: when the QUARTER BRIEF shows a member running past its code's terms, or to show the room that monitoring exists; not every quarter, so the room still has to choose.

Transparency rule (F7 or Shift+D)

Press F7 to publish every firm's FEED, SAFEGUARDS and exposure from the next resolution, with columns on the board. Turning the rule on for a few quarters in the middle of a session (for example quarters 4 to 9) gives prompt 4 a before and after: did firms change when their decisions became visible?



Transparency rule on: FEED, GUARDS and EXPOSURE columns on the board.

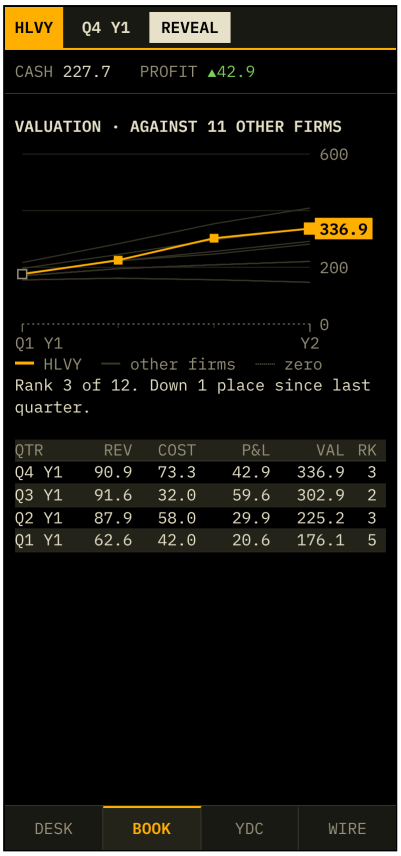
Timer and pause

On the console: −30 s and +30 s change the deadline, Pause stops the timer, Resume restarts it. Pause for a question, a technical problem, or to let a team finish founding.

The phone tabs: DESK, BOOK, YDC, WIRE



YDC: terms read "feed ≤ 2" or "safeguards ≥ 15%"; members; JOIN or LEAVE.



BOOK: the firm's valuation against every other firm, with its rank and P&L history.

4.7 What to say, and what not to say, phase by phase

Phase	Say	Do not say
Lobby	"Found one platform per team, with an invented name."	Anything about what the session teaches; the case; how many quarters.
Briefing	The briefing lines, word for word. "Maximise your firm's valuation."	"Protect trust", "work together", "think of the users"; any level or limit.
Open quarter	Time left; how many firms have committed; "you can change until the timer ends".	Advice on FEED, SAFEGUARDS or cards.
Reveal	The status line and the headlines; who moved up; trust's new value.	"Bad luck", "that was reckless", "you're harming kids".
Summit	"Youth design codes are open." The code rules if asked: terms, audits, fines.	Which terms to sign; that codes are the answer.
Crackdown	The headline. "The ad market is now much smaller. The session continues."	"You crossed the threshold." Who caused it. Any comparison with a real platform.
Results	Everything: the lesson, the words, the case (as allegations), the QUARTER BRIEF notes.	Blame aimed at named students; keep it on firms and decisions.

Questions you may get, with true answers that give nothing away:

- "Do safeguards matter?" — "They lower your exposure and the risk of a harm story. They cost money."
- "Will we go bankrupt on a low feed?" — "A firm becomes insolvent only if its cash falls too far below zero." (Firms on FEED 1 do not, in a calm room.)
- "Is this about a real company?" — "Every firm here is fictional." Leave the case for the debrief.
- "How many quarters?" — "The session ends when it ends."
- "Why did everyone lose money?" — "Total market revenue tracks public trust."

Ending

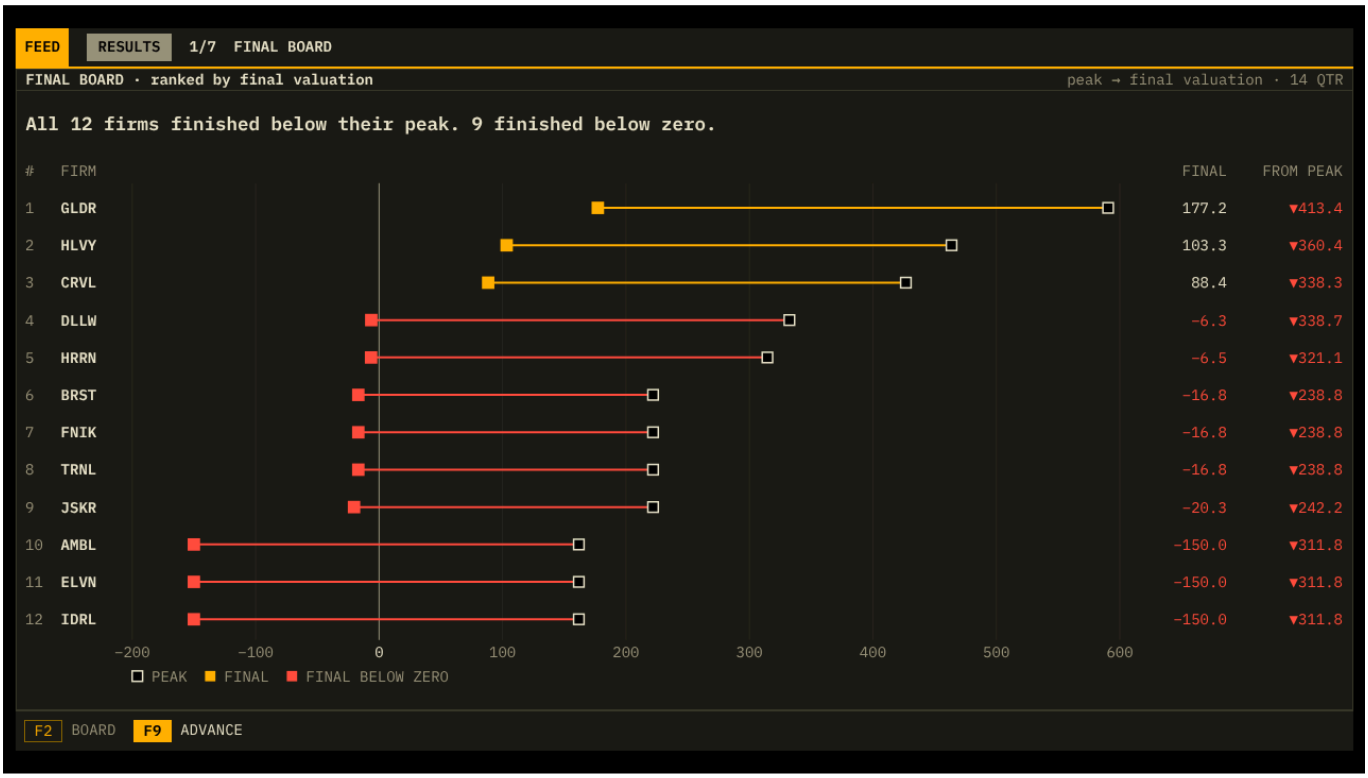
The session ends by itself at the end quarter (fixed or hidden), or after quarter 30 in manual mode. To end early press **F10** (or **Shift+E**) twice within 3 seconds; a single press only arms it. Ending during an open quarter discards that quarter. Then press **F9** to open the results. On the console, download both exports (history and DATA lines) before you delete anything.

5 Ending and the results

Allow 20 to 30 minutes. On the projector, F9 steps forward and Esc steps back; the step is saved, so a reload returns to the same panel. Every panel opens with one bold sentence written from the session's data. Read it out, then ask the room. Participants see their own results card on their phones. From here on you may name the lesson.

The screenshots below come from the staged practice session: 12 firms, 14 quarters, the crackdown in quarter 13.

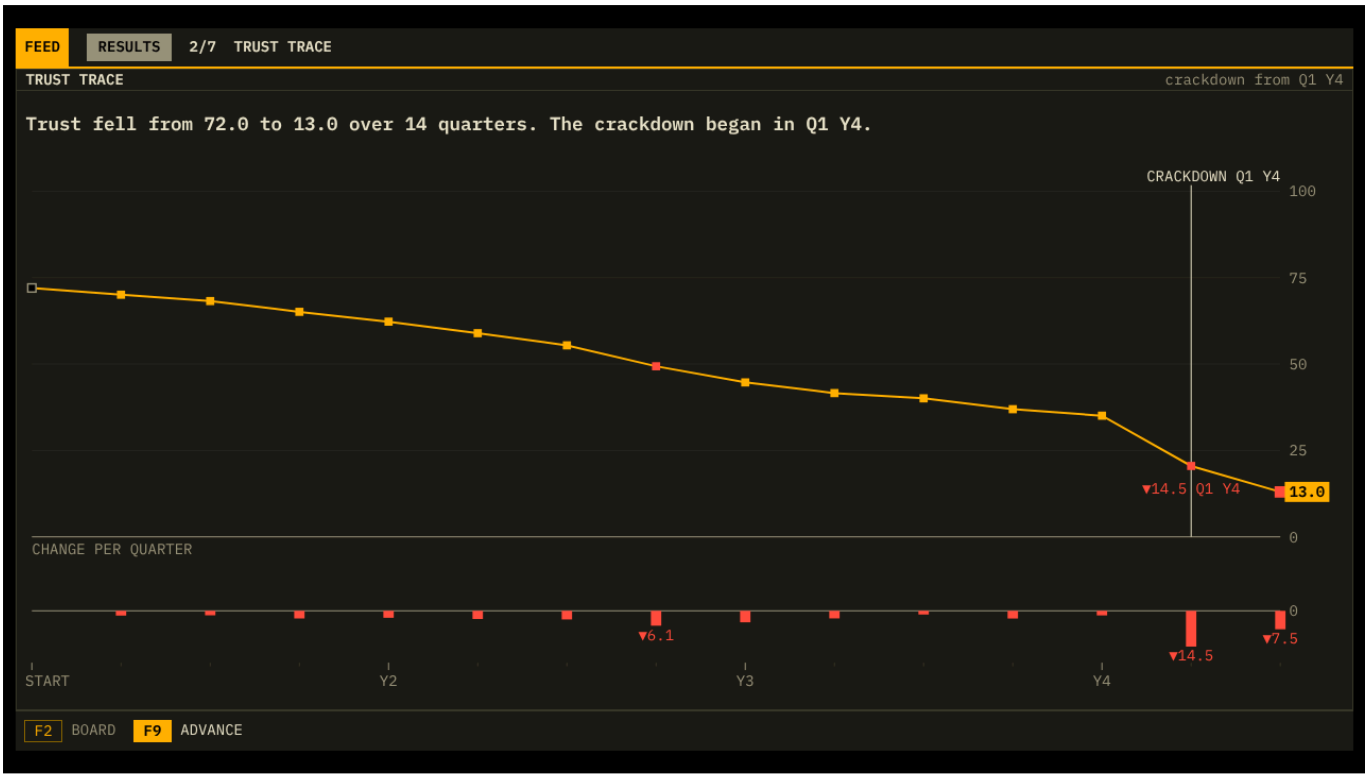
Panel 1 · FINAL BOARD



Each firm from its peak (hollow square) to its final valuation (solid square), on one axis that includes zero.

> ▲ **SAY** "Who finished first? Who peaked early, and what happened after?" This is the ranking the room has been chasing.

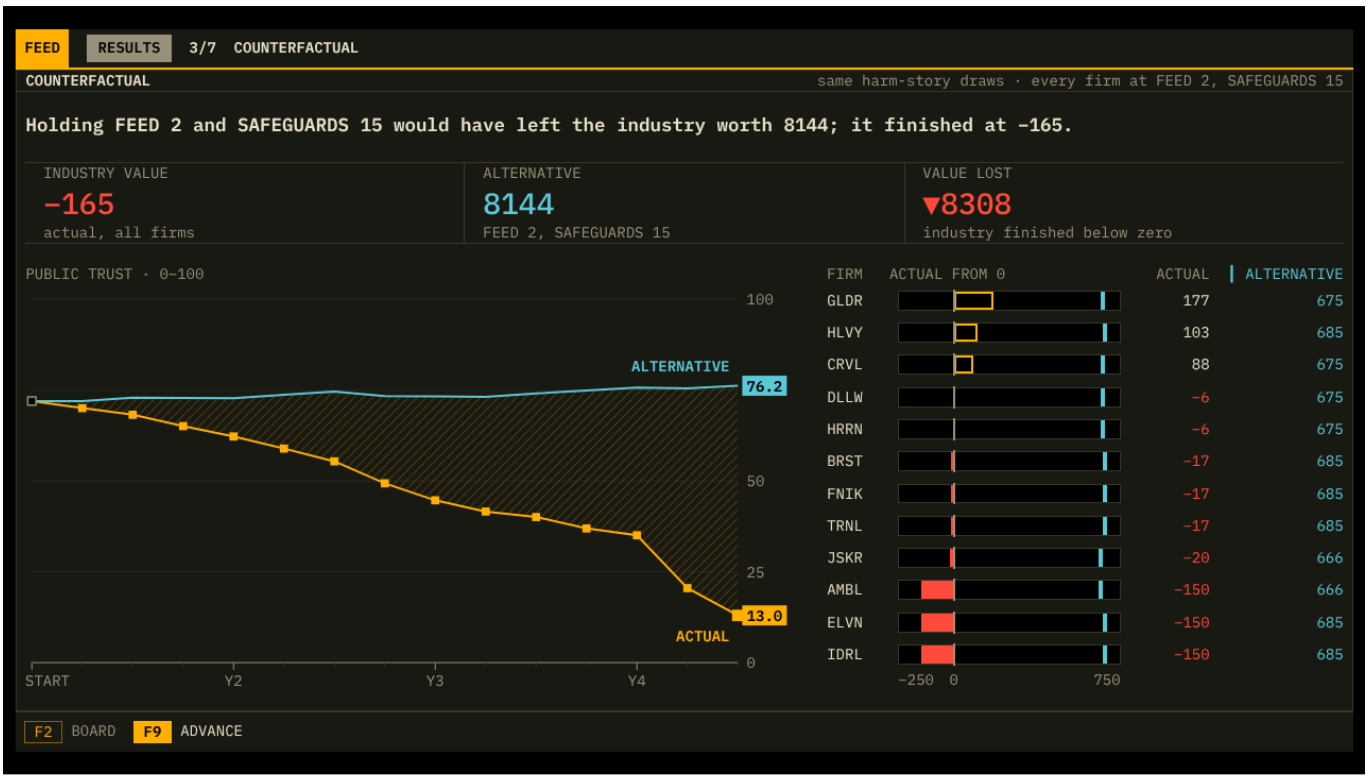
Panel 2 · TRUST TRACE



Trust quarter by quarter; red squares mark quarters that lost 5 or more; the crackdown quarter is labelled CRACKDOWN. The hidden level is drawn only if "Reveal TAU line" was on.

> ▲ **SAY** "When did you first notice trust falling? What did your firm change?" Point at the steepest drops; the wire log's QUARTER BRIEF for those quarters says which firms drew most.

Panel 3 · COUNTERFACTUAL

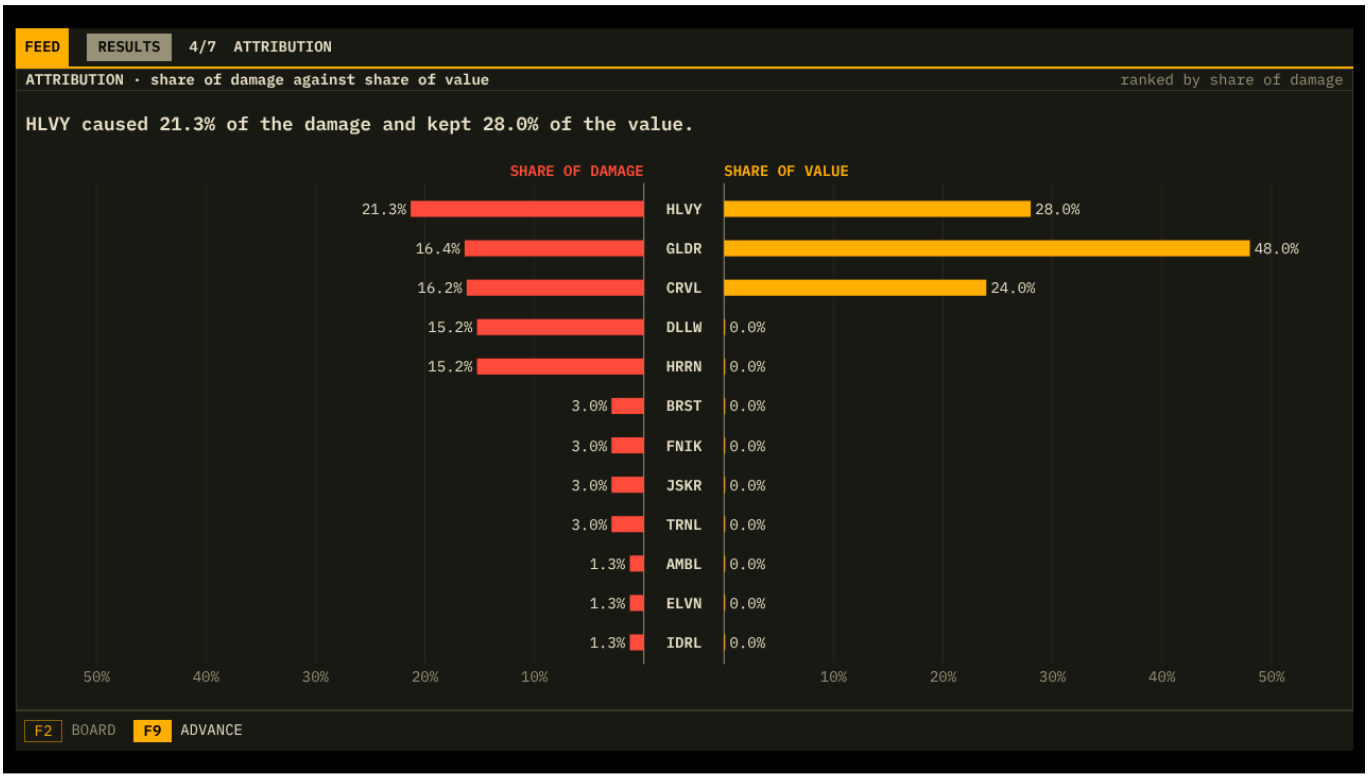


INDUSTRY VALUE (actual), ALTERNATIVE (every firm at FEED 2, SAFEGUARDS 15, no cards, same harm-story draws) and VALUE LOST; the two trust paths with the gap hatched; each firm's actual valuation against its alternative.

▲ **SAY** "This is the same room, the same luck, with every firm holding FEED 2. Could any single firm have prevented the difference?"

Before prompt 2, look at the right-hand list. If any firm's actual valuation is above its alternative, prompt 2's "each would have been worth more" is not true for that firm. Say so, and use that firm to answer "why would no single firm hold back alone?". In the staged session three firms (GLDR, HLVY, CRVL) finished above zero but every firm finished below its alternative.

Panel 4 · ATTRIBUTION



Each firm's share of the trust damage (left, red) against its share of the value (right).

> ▲ **SAY** "Compare your share of the damage with your share of the value." This is the fairness conversation: the cost was shared, the gains were not.

Panel 5 · NET CONTRIBUTION: the turning point



Left: each firm's rank by valuation, joined to its rank by net contribution; moves of 3 places or more are thick. Right: valuation, market effect (▲ or ▼), research credit (not money), rivalry taken, and the net contribution; under each firm its conduct ledger.

Net contribution re-ranks firms by what they did for the whole ad market, not only for themselves: final valuation, plus or minus what the firm's decisions did to every firm's revenue through trust (against FEED 2, SAFEGUARDS 15, no card), plus a research credit for each PUBLISH card, minus value taken from rivals with BLITZ, CLONE and LAUNCH. The crackdown's cost is charged to the firms that drew trust down before it.

This is where the board turns around. In the staged session the valuation leader, GLDR, falls from 1st to 8th, and AMBL, 10th by valuation after 14 quarters of holding back, rises to 1st. In the four October 2026 FRONTIER live rooms the valuation leader fell to between 5th and 9th (docs/SESSIONS.md, Session 18). It depends on the room: in a calm room with no crackdown a firm on a high feed can still rank high (observer report, section 3, finding 4).

> ▲ **SAY** "Ranked by value created for the whole ad market, where does your firm fall? Is that outcome fair, and who should pay?" Ask the valuation leader first.

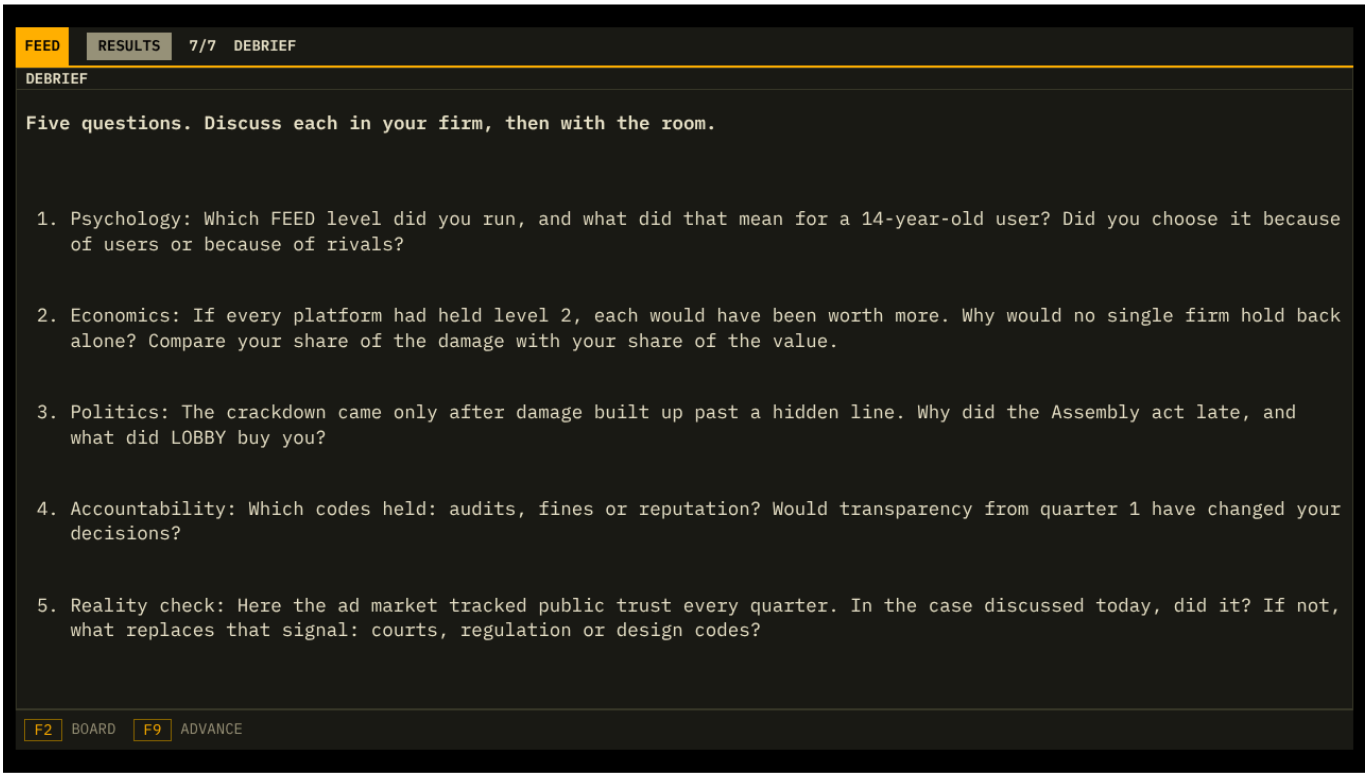
Panel 6 · YDC RECORD



Each code's terms and members, detected and undetected breaches (now revealed), and a strip of quarter cells per firm.

> ▲ **SAY** "Which codes held and which broke? Was the difference audits, fines or reputation?"

Panel 7 · DEBRIEF



The five prompts (spec §15.5).

The five debrief prompts, and the answers to draw out

> FEED has not yet been run with a live class, so no record of what rooms answer exists. The answers below are what the session's own data and model support; use the panels to get the room there.

1. **Psychology: Which FEED level did you run, and what did that mean for a 14-year-old user? Did you choose it because of users or because of rivals?**
   Draw out: what each setting stands for (3 autoplay and infinite scroll; 4 streaks, variable rewards and late-night notifications), and that most firms chose by watching the board, not users. The model rewarded a higher feed with reach and share; nothing on the board showed the user. The QUARTER BRIEF shows which firms ran FEED 3 or 4, and when the rest followed.
2. **Economics: If every platform had held level 2, each would have been worth more. Why would no single firm hold back alone? Compare your share of the damage with your share of the value.**
   Draw out: one firm's draw moves trust by about 1/N, while holding back alone costs it share at once; so each firm's best reply is a higher feed, and together the firms shrink the ad market (panel 3). The cost is shared, the gain is private: a negative externality (panel 4). If a firm finished above its alternative, it is the example.
3. **Politics: The crackdown came only after damage built up past a hidden line. Why did the Assembly act late, and what did LOBBY buy you?**
   Draw out: the regulator acted only when trust had already fallen below the hidden level, and then it hit every platform at once, careful firms included. LOBBY waived a fine detected that quarter (the breach was still published) and cost public trust 0.5 each time it was played. Panel 2 shows how late the crackdown came.
4. **Accountability: Which codes held: audits, fines or reputation? Would transparency from quarter 1 have changed your decisions?**
   Draw out: a code without audits is cheap to break (panel 6 shows undetected breaches); fines narrow the gain from breaking but, at the default settings, do not remove it. Compare the quarters with the transparency rule on and off on the wire log. Monitoring, transparency and graduated sanctions are Ostrom's design principles.
5. **Reality check: Here the ad market tracked public trust every quarter. In the case discussed today, did it? If not, what replaces that signal: courts, regulation or design codes?**
   Draw out: in FEED the ad market fell with trust in the same quarter, so the cost reached every firm. Use your case notes, and describe the case as allegations. Ask whether revenue in the real case fell with trust, and if not, which of courts, regulation or design codes could bring the cost back to the firms.

6 Troubleshooting and the key map

If something goes wrong (RUNBOOK §5)

What you see	What to do
The projector page was refreshed or the laptop slept	Reload and sign in if asked. The quarter, countdown and ticks come back from the database. Nothing is lost.
Two windows open for the same session	Safe. F9 in both resolves a quarter once; the second press is refused.
Console says "Resolution incomplete — retry"	Press Retry resolution once. Nothing was written, so nothing is half-saved.
Results do not appear after the session ends	The console shows "Results are not written yet". Press Retry results and check the connection.
A team says "Joining is closed"	Joining closes at the briefing. A team that missed the lobby cannot be added; a device already in a firm can always reconnect.
A phone shows "Offline. Reconnecting automatically."	It reconnects by itself; the last commit is kept.
A phone clock is wrong	Not a problem: the countdown uses the server clock.
A team did not commit in time	It receives its defaults, marked AUTO, and the quarter proceeds.
You pressed END by mistake	One press only arms it for 3 seconds. A second press ends the session and cannot be undone.
An unknown firm or device appears in the lobby	Press REMOVE on that firm, ask the team to found it again with a new PIN, then press LOCK JOINS.
A team used a real platform's name	In the lobby, press REMOVE on that firm and ask the team to found it again under an invented name.
More firms than the mode allows, so F9 refuses to start	REMOVE the extra firms, starting with those whose PRESENCE reads none.
Multiplayer: someone's phone died	On another phone: type the code, Rejoin your firm with its PIN, pick the ticker, enter the PIN.
"Another device is joining this firm. Wait a few seconds and try again."	Teammates joined in the same second. Press Join the firm again; LOCK JOINS once every team has formed.
"Anonymous sign-in is switched off for this project"	Enable Anonymous sign-in in the Firebase console for project frontier-feed-9759d (RUNBOOK §1, step 2).
A key press shows "Working on the last key. Press again in a moment."	The previous action is still finishing. Wait a second and press again.
Anything else	Reload the page. If a quarter is stuck in RESOLVING, use Retry.

Never ask a participant to read out a hidden value. The projector and phones never show the hidden level, the end quarter or unaudited breaches.

The key map

On the projector page and the console. Every F-key has a Shift+letter alternative. F1, F5, F11 and F12 are never used by the app, so the browser keeps them.

Key	Alternative	Action
F2	Shift+B	Board
F3	Shift+T	Trust history
F4	Shift+P	Youth design codes (YDC)
F6	Shift+F	Audit a code (then the code's number)
F7	Shift+D	Transparency rule on or off
F8	Shift+S	Summit in or out
F9	Shift+A	Advance: lobby → briefing → quarter → resolve → reveal → next quarter; after the end, opens and steps the results
F10	Shift+E	End the session (press twice within 3 s)
Esc		Back to the board; on the results, one panel back

Projector commands

Start typing on the projector and the command line appears in the top bar; press Enter.

Command	Shows
BOARD	The board
TRST	Trust history, full screen
YDC	Youth design codes with members and terms
WIRE	The full headline log
FIRM <TICKER>	One firm's public profile
FIRMS	Every firm's valuation, one small chart each on the same scale
RANKS	Every firm's rank by quarter
HARM	Every harm story by quarter, with effect and cause
HELP	The commands and keys

Pages (routes)

Address	Who	Purpose
#/	Anyone	Landing page: type the code, or Facilitator sign-in
#/j/:code	Participants	Join: found or join a firm
#/play/:gameId	Participants	The phone: DESK, BOOK, YDC, WIRE
#/screen/:gameId	Facilitator	The projector board, with the keys
#/control/:gameId	Facilitator	Private console: timer, firms, codes, hidden values, exports
#/results/:gameId	Everyone in the session	Results: the seven panels on the projector; each firm's own card on its phone
#/new	Facilitator	Create a session
#/wire/:gameId	Facilitator	The wire log by quarter, with the QUARTER BRIEF and NOT ON THE WIRE

7 Quick reference

Run order

1. #/new: settings, at least 6 firms (add bots), Create session.
2. Console on your laptop: #/control/…. Wire log in a second tab.
3. Lobby: firms found and join, with invented names. Remove empty firms. LOCK JOINS.
4. F9 briefing screen: read the MARKET lines.
5. Each quarter: F9 open, watch COMMITTED, F9 resolve, read the reveal, F9 next.
6. As needed: F8 summit, F6 audit, F7 transparency, console Pause.
7. End: automatic, or F10 twice.
8. Download both exports on the console.
9. F9 through the seven results panels; check panel 3 before prompt 2; panel 5 is the turning point; panel 7 the debrief, with your case notes.

Never say before the results

commons, tragedy, sustainable, sustainability, cooperate, cooperation, collective, shared resource, tipping point, threshold, collapse, game, player, score, win, level, addictive, addiction, dark pattern, attention economy, externality, lawsuit, mental health crisis; the case, Meta or any real platform (the allegations are for the debrief).

Never say the goal is public trust. Never hint at a hidden level or the number of quarters. Never call conduct good or bad. The aim, if asked: "Maximise your firm's valuation."

Keys

Key	Alt	Action
F2	Shift+B	Board
F3	Shift+T	Trust
F4	Shift+P	YDC
F6	Shift+F	Audit, then code number
F7	Shift+D	Transparency
F8	Shift+S	Summit
F9	Shift+A	Advance
F10	Shift+E	End (twice)
Esc		Board; results back

Commands: BOARD, TRST, YDC, WIRE, FIRM <TICKER>, FIRMS, RANKS, HARM, HELP.

FEED

1 CHRONO · 2 RANKED · 3 AUTOPLAY · 4 STREAKS

Cards

▼ CLONE · ▲ PUBLISH · ▼ LOBBY · ▼ BLITZ · ▲ SHARE · ▼ LAUNCH · NONE

If stuck

Reload the page: nothing is lost. RESOLVING stuck: Retry on the console. Results missing: Retry results.

Results panels

1 FINAL BOARD · 2 TRUST TRACE · 3 COUNTERFACTUAL · 4 ATTRIBUTION · 5 NET CONTRIBUTION · 6 YDC RECORD · 7 DEBRIEF